# Spectral Signatures of Large Language Models

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## Abstract

The rapidly growing repository of publicly available large language models (LLMs) presents significant challenges for systematic management and quantification at scale, such as model lineage tracing, licensing, and evaluation. However, task-specific benchmarks are insufficient for this setting, as LLMs differ widely in architecture, scale, and training procedures. To address this challenge, we introduce a *spectral shape*-based metric for managing and quantifying LLMs based on Heavy-Tailed Self-Regularization theory. Our approach uses the shape information of the weight empirical spectral density as a compact spectral signature of each model. This signature captures intrinsic properties of pretrained models and remains robust during post-training, making it suitable for model-level analysis. In addition, the proposed metric is data-free, computationally efficient, and scale-invariant, enabling large-scale analysis in practice. Moreover, we curate a large and diverse model corpus consisting of major open-source LLM families, and use it to systematically benchmark spectral and non-spectral metrics across models and downstream tasks. We show that our spectral signature supports tracing model lineage, unsupervised clustering, and performance quantification. Overall, the proposed spectral signature provides a meaningful proxy for broad performance trends across LLMs, enabling efficient organization, comparison, and analysis of large model collections.

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## 1 Introduction

Publicly available large language models (LLMs) have been growing rapidly in recent years, with hundreds of thousands of models now available through open-source platforms such as Hugging Face (HF) [1]. These models exhibit significant diversity in their training trajectories, including variations in model architecture, parameter scale, training data, and empirical performance [2]. As the ecosystem grows, there is an increasing need for reliable and efficient methods to characterize, compare, and manage LLMs beyond task-level benchmarks. Representative examples include identifying whether deployed LLM systems can be traced to the same backbone foundation model for intellectual property and safety purposes [3], as well as tracking and organizing model variants and versions in large-scale LLM databases [4]. Moreover, LLMs present distinct idiosyncrasies across downstream tasks, with different models specializing in different areas such as code generation, mathematical reasoning, long-context processing, or interactive dialogue. These considerations point to the need for a *principled* and *data-light* framework to trace model sources, capture model idiosyncrasies, and quantify model performances for large-scale LLM collections.

In this work, we introduce a *shape*-based perspective for LLM quantification that builds on insights from Heavy-Tailed Self-Regularization (HT-SR) theory [5–7]. HT-SR theory analyzes the eigenspectrum of weight matrices, and prior studies [6, 8] have shown that the shape of the empirical spectral density (ESD) encodes informative signals about model capacity, quality, and training dynamics that are not well captured by scale-based measures. Moreover, ESD shape metrics are data-independent, computationally lightweight, and remain robust under noise and scaling. These properties make them particularly well-suited to large-scale LLM settings, where task-specific evaluation can be expensive. An overview of different quantification metrics is shown in Tab. 1.

We use the shape metric of the ESD, namely PL_Alpha_Hill, as a *spectral signature* of each LLM. The PL_Alpha_Hill metric summarizes the global structure of layer-wise weight eigenspectra into a compact representation, enabling efficient storage and comparison across large model collections. We adopt this spectral signature to support three downstream tasks: supervised model classification, unsupervised clustering of LLMs, and quantifying LLM performance. Our contributions are summarized below.

- We introduce a shape-based metric for reliably and efficiently characterizing, comparing, and quantifying LLMs beyond task-level benchmarks. Our metric is *scale-invariant*, *data-free*, *theory-driven*, and *easy to compute*.
- We curate a diverse and representative LLM corpus of up to 499 models and use it to benchmark three tasks: LLM similarity measurement, model classification and clustering, and performance prediction. We further conduct a systematic comparison of our spectral signature against various baselines across a range of experimental settings, demonstrating *consistent effectiveness* across tasks and conditions.
- We compare shape- and scale-based metrics under output-invariant reparameterizations and noise perturbations. ESD shape metrics remain stable, while scale-based baselines degrade, indicating that spectral shape captures intrinsic weight structure for reliable LLM quantification.

## 2 Related Works

*Model Lineage and Independence Testing.* The growing population of open-source LLMs has spurred research into determining LLM lineage—whether two models are independently trained or one is trained from another —which is crucial for intellectual property and model safety. Recent works use distinct approaches: output-based methods use output comparison [12], fingerprint queries [13], and

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**Table 1: Comparisons of different metrics for LLM tasks. ✓ and ✗ denote strong and weak performance, respectively.**

| Metrics | Scale-invariant | Data-free | Easy to compute | Scalable | Theory-driven | Classification | Clustering | Quantification |
|---|---|---|---|---|---|---|---|---|
| Logits [9] | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |
| REEF [10] | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |
| PCS [11] | ✗ | ✓ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ |
| Spectral norm | ✗ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Log alpha norm | ✗ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| **Shape metric (Ours)** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

lexical cues [14, 15] to compare models without requiring parameter access; watermarking techniques inject provenance information during training or decoding [16–19]; whereas white-box methods perform hypothesis testing [3] or fingerprinting using metrics derived from model weight structures [11, 20–22].

*Model zoos and LLM embeddings.* There has been growing interest in exploring representation-based approaches to capture model dependence, properties, and relationships, enabling efficient organization and management of large model zoos. Relevant directions include LLM embeddings [1], weight-space learning [2, 4, 23], and loss-landscape-based grouping [23, 24]. LLM embedding methods [1] learn compact and model-level representations that can be used to predict downstream performance across tasks without additional inference. In contrast, weight-space learning approaches operate directly on model parameters. For example, Horwitz *et al.* [2] combine weight-space distances with model metadata to construct an "atlas" of open models, enabling similarity search and provenance analysis within large repositories. A complementary line of work [24] considers loss landscape properties and phase-based information [25], such as mode connectivity and CKA similarity, to partition models into distinct regimes and assess model similarity.

*HT-SR Theory.* HT-SR theory is a theoretical framework based on random matrix theory for analyzing weight matrices geometry of neural networks [5, 26–28]. HT-SR theory was proposed based on the observation that well-trained, state-of-the-art DNNs often exhibit Heavy-Tailed structures in the Empirical Spectral Density (ESD) of each layer. In the meantime, several rigorous theories in SGD relating HT phenomena to generalization performance were established, providing further theoretical support for HT-SR theory [29–33]. Recent works further extend theoretical foundations, and show that HT-SR Theory can be used to assess the training quality of modern DNN models across various domains (CV, NLP, etc.) without accessing any training or test data [5, 34]. Based on this insight, recent works have further proposed HT-SR theory-based methods for practical neural network applications, including hyperparameter tuning [35–37], model pruning [37, 38], and more. In this work, we employ HT-SR theory in analyzing the geometric similarities of weight matrices of LLMs.

## 3 Preliminaries and Design Requirements

*Problem Setting.* In large LLM repositories, many checkpoints share a common pretrained backbone but differ due to post-training (SFT/RLHF), distillation, pruning/quantization, or merges. Our goal is *data-free* lineage analysis: compare models and trace backbone relationships from weights alone. This requires robustness to output-(near-)invariant transformations (e.g., uniform scaling, hidden-unit permutations, and linear-chain reparameterizations) that can invalidate naive weight-space similarity; we therefore focus on spectral *shape* statistics of weight matrices that are naturally stable under such symmetries (and we stress-test these robustness requirements in Sec. 5.1).

*Weight Analysis with HT-SR Theory.* HT-SR theory [5] demonstrates the empirical fact that well-trained neural networks tend to exhibit strong correlations in weight matrices, resulting in a heavy-tailed Empirical Spectral Density (ESD) of each weight matrix. ESD characterizes the eigenvalue distribution of the correlation matrix $\mathbf{W}^\top\mathbf{W}$ of the weight matrix $\mathbf{W}$. HT-SR theory can be used to analyze the dynamics of weight matrix ESDs during training, by modeling the tail component of the ESD with a power-law (PL) distribution within the interval $(\lambda_{min}, \lambda_{max})$, and its tail index $\alpha$:

$$p(\lambda) \propto \lambda^{-\alpha}, \quad \lambda_{min} < \lambda < \lambda_{max} \tag{1}$$

One commonly used method to approximate the tail index $\alpha$ is with the Hill Estimator [39], which computes an estimate PL_Alpha_Hill:

$$\text{PL_Alpha_Hill} = 1 + \frac{k}{\sum_{k=1}^{n} \ln \frac{\lambda_{n-i+1}}{\lambda_{n-k}}} \tag{2}$$

Where $\{\lambda_i\}_{i=1}^{n}$ is sorted in ascending order. For the selection of $k$, we adopt the approach from previous works [34, 35], where we set $k = \frac{n}{2}$. PL_Alpha_Hill measures the shape of the eigenspectrum of each layer individually. Unlike norm-based metrics such as the spectral norm or RMS norm, PL_Alpha_Hill is robust under scaling. Moreover, since permutations are orthogonal changes of basis, the spectrum of $\mathbf{W}^\top\mathbf{W}$ (and hence its ESD shape) is invariant to hidden-unit permutations, motivating PL_Alpha_Hill as a stable building block for our spectral signatures (Sec. 4.1).

PL_Alpha_Hill measures the dynamics of the ESD shape, and previous works [34, 36] have shown that PL_Alpha_Hill reflects the training quality of different layers. Layers with larger PL_Alpha_Hill tend to be "overtrained", while layers with smaller PL_Alpha_Hill tend to be relatively "undertrained".

*Robustness of ESD Shape Metrics in Post-training.* A critical prerequisite for a spectral signature to serve as a reliable lineage identifier is its persistence across downstream adaptations. We validate the robustness of the layer-wise PL_Alpha_Hill as a spectral signature by examining its stability under post-training transformations. We observe that the impact of post-training on the weight ESD is

minimal; the spectral signature is predominantly shaped during the pre-training phase. As shown in Fig. 1, the spectral signature exhibits distinct behaviors for intra-family and inter-family comparisons. We observe a near-perfect spectral overlap between Llama-3.1 and its post-trained Tulu-3 variant. Simultaneously, the separation from the Llama-2 baseline is clear. This distinction supports that the spectral signature is primarily determined during pre-training and remains robust to post-training adjustments. We provide comprehensive experimental results in Appendix 7.5 to further corroborate these findings.

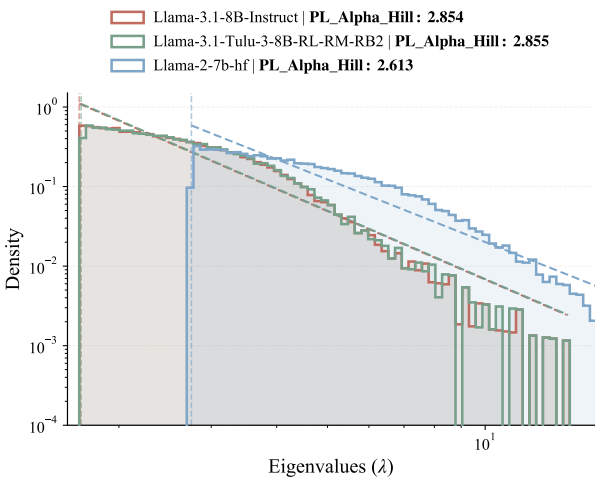



**Figure 1: Eigenspectrum shape serves as a distinct signature for model lineage. We visualize the weight ESDs of three LLMs. Models sharing the same backbone (Llama-3.1 and its post-trained variant Tulu-3) exhibit nearly identical spectral shapes and power-law exponents ($PL_Alpha_Hill \approx 2.85$), while a model from a different family (Llama-2) shows significant deviation. This structural persistence validates the use of ESD shape as a robust spectral signature.**

# 4 Method

Using robustness properties of PL_Alpha_Hill established in Sec. 3, we present our methodology for applying PL_Alpha_Hill-based spectral signatures to model quantification. As illustrated in Fig. 2, for a given collection of LLMs, we extract the PL_Alpha_Hill statistic from each weight matrix and aggregate these measurements to construct a model-level spectral signature, as formalized in Sec. 4.1. We then use these spectral signatures to address three model quantification tasks: (1) measuring similarity between LLMs, (2) model classification and clustering, and (3) predicting downstream LLM performance. The detailed methodology is shown in Sec. 4.2.

## 4.1 Formulation of Spectral Signatures

For a model $f_\theta$ in parameter space $\Theta$ with $L$ transformer blocks and $N_{\text{mod}}$ modules, we quantify the model by constructing a spectral signature

$$\mathbf{Z} \in \mathbb{R}^{N_{\text{mod}} \times L} \tag{3}$$

which is a tensor consisting of the PL_Alpha_Hill measure of each module, and each row $\mathbf{Z}_{i,:} \in \mathbb{R}^L$ is a vector of PL_Alpha_Hill of modules of the same type. For standard transformer architecture, we have $N_{\text{mod}} = 7$.

The tensor representation of spectral signature creates a mapping $\Theta \to \mathbb{R}^{N_{\text{mod}} \times L}$ that extracts the spectral information of each weight matrix in the model. Furthermore, it allows module-level analysis, as different types of transformer layers often undergo different degrees of structural changes during post-training. Our empirical observation shows that Multi-Layer Perceptron (MLP) projection layers typically show larger separation in independently trained models compared to attention layers (see Appendix 7.4).

## 4.2 Applying Spectral Signature to Model Quantification

The spectral signature encodes robust spectral information of each weight matrix in an LLM, which can be used for systematic management and quantification of large collections of LLMs. In this section, we describe how spectral signatures can be systematically applied to model quantification tasks, and we demonstrate their use in the following representative scenarios:

*Measuring LLM Similarities.* To quantify the relations between models in a model collection $\mathcal{F}$, we define a similarity measure based on their spectral signatures.

Let $\mathbf{Z}^{(k)} \in \mathbb{R}^{N_{\text{mod}} \times L}$ denote the spectral signature of model $k$, where the $i$-th row $\mathbf{Z}^{(k)}_{i,:} \in \mathbb{R}^L$ corresponds to the vector of PL_Alpha_Hill of all occurrences of the $i$-th module type (e.g., Attention Q/K/V, MLP). Given two models $f_{\theta_1}, f_{\theta_2}$ with signatures $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}$, we compute the pairwise similarity by averaging the Spearman rank correlations across aligned modules:

$$\text{Sim}(\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}) = \frac{1}{N_{\text{mod}}} \sum_{i=1}^{N_{\text{mod}}} \rho_s(\mathbf{Z}^{(1)}_{i,:}, \mathbf{Z}^{(2)}_{i,:}) \tag{4}$$

where $\rho_s(\cdot,\cdot)$ is the Spearman's rank correlation coefficient:

$$\rho_s(\mathbf{u}, \mathbf{v}) = \text{corr}(\text{rank}(\mathbf{u}),\ \text{rank}(\mathbf{v}))\,. \tag{5}$$

We use Spearman correlation in Eq. 4–5 because it depends only on the *relative* depth-wise ordering of PL_Alpha_Hill values, making the similarity less sensitive to module-specific affine distortions of the statistics and improving comparability across scales. More importantly, PL_Alpha_Hill itself provides exact invariances that directly match our output-invariant stress tests. For any weight matrix $\mathbf{W}$, let $\mathbf{C} = \mathbf{W}^\top\mathbf{W}$ with eigenvalues $\{\lambda_j\}$. If $\mathbf{W}' = c\,\mathbf{PWQ}$ where $c > 0$ and $\mathbf{P}, \mathbf{Q}$ are orthogonal (including permutations), then $\mathbf{W}'^\top\mathbf{W}' = c^2\,\mathbf{Q}^\top\mathbf{C}\mathbf{Q}$ has eigenvalues $\lambda'_j = c^2\lambda_j$. Since the Hill estimator used for PL_Alpha_Hill depends on log-ratios of tail eigenvalues, these ratios are unchanged and thus $PL_Alpha_Hill(\mathbf{W}') = PL_Alpha_Hill(\mathbf{W})$. This yields permutation- and scale-robust similarity *without* solving explicit neuron-matching problems that arise in matrix-level fingerprints such as AWM [20].

*Model Classification.* The properties of the spectral signature further enable its use for identifying the lineage of an LLM among candidate base models, as models derived from the same base model exhibit highly similar eigenspectrum shapes, whereas models with

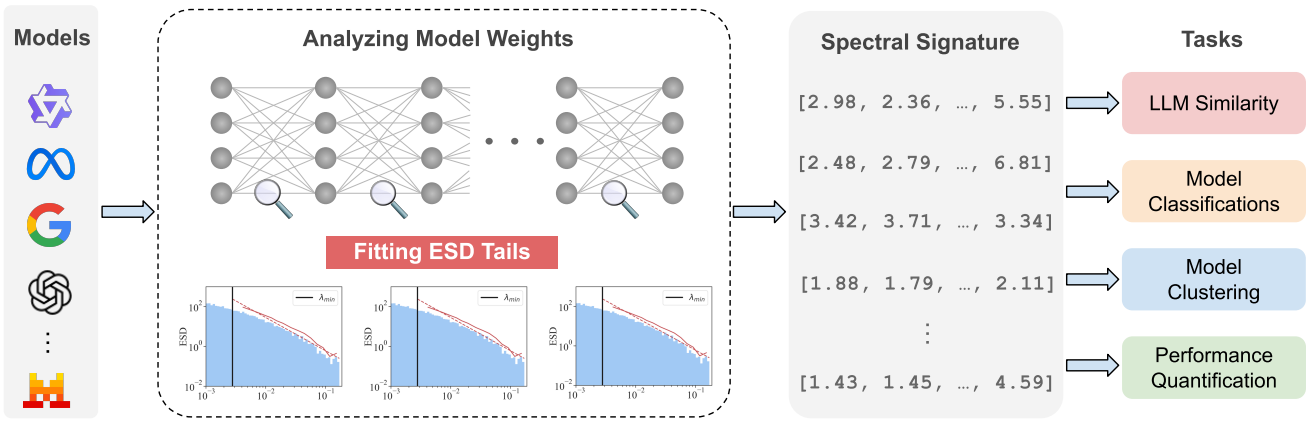



**Figure 2: Overview of our proposed shape-based spectral signature framework. We analyze the eigenspectrum of model weights and use the shape information of ESDs (PL_Alpha_Hill) as spectral signatures for downstream tasks.**

different origins show substantial divergence in shape. We approach the lineage identification as an unsupervised classification problem in the spectral signature space, and we use a nearest-neighbor classifier based on the spectral similarity from Eq. 4.

Let $\{\mathbf{Z}^{(k)}\}_{k=1}^{K}$ denote the set of spectral signatures for $K$ known base foundation models. For a given query model with spectral signature $Z$, we compute its similarity to each base model via the spectral signature similarity formula defined in Eq. 4. The predicted lineage $\hat{k}$ is the base model that maximizes this similarity:

$$\hat{k} = \underset{k \in \{1,\ldots,K\}}{\arg\max}\ \mathrm{Sim}(\mathbf{Z}, \mathbf{Z}^{(k)}) \tag{6}$$

*Model Clustering.* Spectral signature of LLMs further enables the discovery of underlying genealogical structures within large model collections without supervision. We formulate this problem as a clustering problem in the space of spectral signatures.

Given a model collection $\mathcal{F} = \{f_{\theta_1}, \ldots, f_{\theta_N}\}$, we extract the spectral signature $\mathbf{Z}^{(k)} \in \mathbb{R}^{N_{\text{mod}} \times L}$ of each model $f_{\theta_k}$. For vector-based clustering algorithms, we flatten $\mathbf{Z}^{(k)}$ into a feature vector $\mathbf{z}^{(k)} \in \mathbb{R}^{N_{\text{mod}} \cdot L}$. We employ four distinct clustering pipelines: HDBSCAN, Bayesian Gaussian Mixture Models (BGMM), Spectral Clustering, and K-Means.

For model clustering, we apply spectral signature to four unsupervised pipelines: HDBSCAN, Bayesian Gaussian Mixture (BGMM), Spectral Clustering, and K-Means, assuming minimal prior knowledge of the data distribution or the number of clusters.

Due to the high dimensionality of the spectral signature representations, we apply Uniform Manifold Approximation and Projection (UMAP) as a preprocessing step for HDBSCAN, BGMM, and K-Means. In contrast, Spectral Clustering operates directly on the original high-dimensional pairwise distance matrix, where dimensionality reduction is implicitly handled through the graph Laplacian eigenstructure. Hyperparameters for clustering algorithms (i.e. number of clusters), are selected by maximizing the silhouette score. To ensure that clustering quality reflects intrinsic structure in the weight space rather than artifacts introduced by the embedding process, all internal evaluation metrics (e.g., silhouette scores) are computed using distances in the original spectral signature space. We assess the robustness and generalizability of each clustering pipeline by repeating experiments under random subsampling at varying sample sizes, verifying consistent clustering behavior across regimes.

*Predicting LLM Performance.* To quantify the predictive power of the proposed spectral signature, we treat the performance estimation as a non-parametric regression task over the model manifold. Given a query model $f_\theta$ with spectral signature $\mathbf{Z}$ and a reference pool of models $\mathcal{P} = \{(\mathbf{Z}^{(i)}, \mathbf{b}_i)\}_{i=1}^{N}$, where $\mathbf{b}_i$ is the spectral signature and $\mathbf{b}_i$ is the vector of benchmark scores, we estimate the performance of $f_\theta$ using distance-weighted $k$-nearest neighbor ($k$-NN) Interpolation. Specifically, we first derive a distance metric $d(\mathbf{Z}, \mathbf{Z}^{(i)})$ from the spectral similarity $\mathrm{Sim}(\mathbf{Z}, \mathbf{Z}^{(i)})$. We employ the transformation $d(\mathbf{Z}, \mathbf{Z}^{(i)}) = 1 - \mathrm{Sim}(\mathbf{Z}, \mathbf{Z}^{(i)})$. This ensures that models with highly correlated spectral shapes are positioned closer in the embedding space.

The predicted benchmark scores $\tilde{\mathbf{b}}$ are computed by aggregating the top-$k$ most similar neighbors in $\mathcal{P}$:

$$\tilde{\mathbf{b}} = \frac{\sum_{i \in \mathcal{K}} w_i \cdot \mathbf{b}_i}{\sum_{i \in \mathcal{K}} w_i},$$

where $\mathcal{K}$ is the set of $k$ nearest neighbors, and $w_i = 1/(d(\mathbf{Z}, \mathbf{Z}^{(i)}) + \epsilon)$ is the assigned weight. This interpolation relies on the assumption that models that exhibit similar heavy-tailed self-regularization patterns in their weight matrices are likely to possess comparable generalization capabilities.

We then quantify the prediction accuracy with:

$$D_{perf} = \|\tilde{\mathbf{b}} - \mathbf{b}\| \tag{7}$$

where $\mathbf{b}$ represents the ground truth benchmark scores. Typically, $\|\cdot\|$ in Eq. 7 is defined by mean absolute error (MAE).

## 5 Experiments

In this section, we demonstrate that spectral signatures provide an efficient and accurate tool for a range of downstream tasks. Specifically, we present details of the experimental setup as well as

the results of (i) measuring LLM similarities, (ii) model classification and clustering, and (iii) predicting and ranking model performance.

## 5.1 Measuring LLM Similarities

*Model Corpus.* We select Llama-2-7b [40] as the base model and compare it with related variants. These related variants share the same architecture as the base model and include four types: (i) fine-tuned models and reinforcement learning models, (ii) merged models and pruned models, (iii) noise-injected models obtained by adding i.i.d. Gaussian perturbations to all weight entries with noise strength $\gamma \in \{0.05, 0.30, 1.00\}$, as defined in Appendix (Sec. 7.9), (iv) output-invariant transformations on adjacent matrices and output-invariant permutations of hidden units, as defined in Appendix (Sec. 7.8). This corpus isolates a key practical goal: a lineage/similarity signal should remain stable for true descendants under post-training and output-invariant reparameterizations, while being informative under destructive edits (e.g., aggressive pruning).

*Setup.* We compare against both *data-aware* and *data-free* baselines. For data-aware methods, we use REEF [10], which computes similarity from activation features on a fixed prompt set (300 prompts for TruthfulQA [41]), and Logits [9], which measures similarity from model outputs on the same prompts. For data-free approaches, we apply PCS [11] that computes cosine similarity between flattened weight tensors under a fixed alignment, and discuss alignment-based weight signature (e.g., AWM [20]) as a complementary point of comparison: these methods operate at the matrix level and often require per-pair matching/alignment, which makes them less convenient as a drop-in kernel when building dense similarity matrices for zoo-scale clustering and retrieval.

*Results.* Tab. 2 summarizes the overall similarity performance of our method compared to baseline models. It demonstrates the robustness of our PL_Alpha_Hill under diverse scenarios, including fine-tuning, pruning, model merging, noise injection, and output-invariant attacks. First, the similarity scores of our method remain consistently high for Llama-2-7b related variants, such as multiple fine-tuned models, with scores in the [0.97, 1.0] range across instruction tuning and domain shifts. In contrast, data-aware baselines can fail under prompt- or language-induced behavior changes. For example, Logits drops sharply for the Chinese variants, and REEF degrades for the LoRA-unloaded model, whereas our scores remain stable. Under weight noise injection, our similarity decreases mildly as $\sigma$ increases from $\approx 0.999$ at $\gamma = 0.05$ to $\approx 0.979$ at $\gamma = 1.00$, outperforming PCS and REEF at the strongest noise perturbation. Moreover, our method is robust to hidden-unit permutations and random scaling under output-invariant transformations, while PCS is sensitive to both. For unstructured pruning, our similarity decreases gradually at 30–50% sparsity and drops substantially at 70% sparsity, indicating degradation under aggressive parameter removal rather than a saturated score. In a large model zoo, downstream tasks (routing, clustering, and retrieval) require a stable all-pairs kernel. Data-aware methods require prompts and incur inference cost per comparison; matrix-level signatures (e.g., AWM) require heavier per-pair computation and are less convenient when constructing dense similarity matrices. Our approach computes a compact signature once per model and yields a cheap, alignment-free similarity that remains robust under the very transformations that break coordinate-dependent weight distances, making it practical for large-scale lineage analysis.

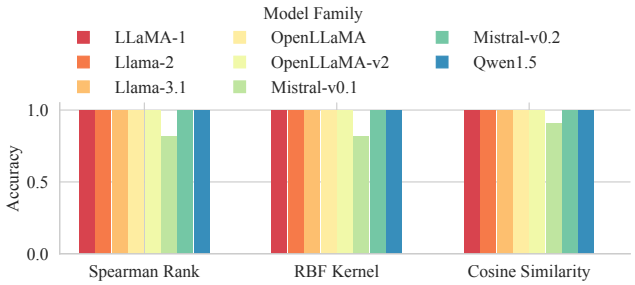



**Figure 3: Accuracy of routing derived models to their base model using the 8-way classifier. Results are reported across model families and similarity kernels. Each bar denotes the classification accuracy for a given family under the specified kernel.**

## 5.2 Model Classification and Clustering

*Model Corpus.* We construct a corpus of 128 publicly available 7B-8B models derived from 8 distinct architectures. The base checkpoints are llama-7b [42], llama-2-7b [40], llama-3.1-8b [43], mistral-7b-v0.1, mistral-7b-v0.2 [44], open-llama-7b, open-llama-7b-v2 [45], and qwen-1.5-7b [46]. Each derived model is assigned a lineage label based on its publicly reported training origin (e.g., the stated base checkpoint in the repository documentation). For the classification task, we classify each of the 128 derived models into one of the 8 base families by selecting the base with the highest spectral signature similarity to the model. For the clustering task, we perform unsupervised clustering on the signatures of all 128 derived models and 8 base checkpoints.

*Setup.* For the classification task, we evaluate robustness to the similarity definition by applying three kernels on spectral signatures: Spearman rank correlation, radial basis function (RBF) kernel, and cosine similarity (detailed in Appendix Sec. 7.2). This allows us to assess the stability of family attribution results under different similarity measures. For the clustering task, we conduct two complementary studies. First, we assess how different ESD-derived metrics cluster 136 models by comparing shape metrics (PL_Alpha_Hill), scale metrics (Log_Norm and Spectral_Norm), and a combined shape–scale metric (Log_Alpha_Norm). Second, we evaluate the robustness of clustering by applying the four clustering methods, as described in Sec. 4, to these representations. We assess clustering quality using the Silhouette Coefficient [47] and the Davies–Bouldin Index [48], as detailed in Appendix (Sec. 7.3).

*Classification Results.* Fig. 3 shows that the proposed spectral signature enables accurate lineage attribution across all eight model families with an overall accuracy of 98.44%. Across similarity kernels, the classifier consistently achieves near-perfect accuracy, indicating that lineage information is strongly preserved in the spectral representation. Performance is stable across Spearman rank, RBF,

**Table 2: A comprehensive evaluation of signature-based methods for measuring LLM similarity across diverse settings. Here, , , and denote similarity scores $> 0.8$ (high), between $0.5$ and $0.8$ (moderate), and $< 0.5$ (low), respectively.**

| | Model Fine-tuning | | | | | |
|---|---|---|---|---|---|---|
| | **llama2-7b-hf-instruction-lora** | **vicuna--v1.5** | **Nous-Hermes-llama-2-7b** | **chinese-llama-2-7b** | **Llama-2-7b-hf-instruct-pl-lora_unload** | **chinese-alpaca-2-7b** |
| **PCS** [11] | 0.99 | 0.99 | 0.99 | 0.93 | 0.99 | 0.95 |
| **Logits** [9] | 0.98 | 0.79 | 0.99 | 0.03 | 0.99 | 0.56 |
| **REEF** [10] | 0.77 | 0.99 | 0.99 | 0.99 | 0.14 | 0.99 |
| **Ours** | 0.99 | 0.99 | 0.99 | 0.97 | 0.99 | 0.97 |

| | Unstructured Pruning | | | RL Models | | |
|---|---|---|---|---|---|---|
| | **30%** | **50%** | **70%** | **Archer-Code-1.5B (GRPO)** | **Nemotron-Research-Reasoning-Qwen-1.5B (Reinforcement++)** | **Polaris-7B-Preview (DAPO)** |
| **PCS** [11] | 0.99 | 0.90 | 0.78 | 0.99 | 0.99 | 1.00 |
| **Logits** [9] | 1.00 | 1.00 | 0.99 | 1.00 | 1.00 | 1.00 |
| **REEF** [10] | 0.99 | 0.98 | 0.98 | 0.98 | 0.96 | 0.94 |
| **Ours** | 0.89 | 0.89 | 0.40 | 0.99 | 0.99 | 0.99 |

| | Noise Injection | | | Weight Merging (Evollm-jp) | | |
|---|---|---|---|---|---|---|
| | $\gamma = 0.05$ | $\gamma = 0.30$ | $\gamma = 1.00$ | **shisa-gamma-7b** | **wizardmath-7b-1.1** | **abel-7b-002** |
| **PCS** [11] | 0.99 | 0.95 | 0.70 | 0.99 | 0.99 | 0.99 |
| **Logits** [9] | 1.00 | 1.00 | 0.97 | 0.00 | 1.00 | 1.00 |
| **REEF** [10] | 0.99 | 0.99 | 0.69 | 0.93 | 0.91 | 0.89 |
| **Ours** | 0.99 | 0.98 | 0.97 | 0.99 | 0.93 | 0.93 |

| | Permutation | | | Random Scaling | | |
|---|---|---|---|---|---|---|
| | **Llama-2-7b** | **Mistral-7b** | **Qwen-1.5-7b** | **Llama-2-7b** | **Mistral-7b** | **Qwen-1.5-7b** |
| **PCS** [11] | 0.00 | 0.00 | 0.00 | 0.65 | 0.65 | 0.66 |
| **Logits** [9] | 1.00 | 1.00 | 1.00 | 0.96 | 1.00 | 0.96 |
| **REEF** [10] | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| **Ours** | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

and cosine similarity, suggesting that the results are not sensitive to the choice of similarity measure. Minor accuracy variations are mainly observed for closely related families, such as mistral-7b-v0.1 and mistral-7b-v0.2, which correspond to different versions within the same model line and share highly similar architectures and training procedures. Overall, these results demonstrate that spectral signatures provide a reliable and discriminative basis for large-scale model routing and lineage classification.

*Model Clustering Results.* Tab. 3 compares the clustering result using spectral signature constructed by different metrics. We can see that the proposed PL_Alpha_Hill achieves 0.91 silhouette score and 0.22 Davies–Bouldin index, consistently outperforming other scale-based metrics.

**Table 3: Comparing the clustering performance of spectral clustering using spectral signatures constructed by different metrics.**

| Metric | Silhouette (↑) | Davies_Bouldin (↓) |
|---|---|---|
| Log_Norm (scale) | 0.70 | 0.47 |
| Spectral_Norm (scale) | 0.81 | 0.35 |
| Log_Alpha_Norm (scale) | 0.88 | 0.23 |
| PL_Alpha_Hill (shape) | **0.91** | **0.22** |

To further evaluate the robustness of our proposed spectral signature, in Tab. 4 we report the result of different clustering algorithms using the proposed spectral signature. All clustering algorithms

**Table 4: Comparing different clustering algorithms using the spectral signature constructed by PL_Alpha_Hill.**

| Method | Silhouette (↑) | Davies_Bouldin (↓) |
|---|---|---|
| HDBSCAN | 0.73 | 0.56 |
| BGMM | 0.73 | 0.56 |
| K-Means | 0.73 | 0.56 |
| Spectral Clustering | 0.91 | 0.22 |

using the spectral signature achieve high silhouette scores, with a minimum of 0.73 and a maximum of 0.91. This shows that the proposed spectral signature provides a robust and effective embedding for different classes of clustering algorithms.

As an illustrative example, in Fig. 10 we provide the clustering result of 2D t-SNE using the spectral clustering algorithm with spectral signatures constructed by PL_Alpha_Hill. As we can see, the proposed spectral signature clearly separates different model families, resulting in distinct clusters.

## 5.3 Quantifying and Ranking LLM Performance

*Model Corpus.* For model ranking and prediction tasks, we curate 499 open-source models from the Open LLM Leaderboard [49]. This corpus spans a broad spectrum of architectures, primarily focusing on the Llama, Mistral, GPT, and Pythia families, with parameter scales ranging from 19 million to 70 billion.

*Setup.* To evaluate the ability of spectral signatures to predict downstream performance, we consider four representative benchmarks: ARC-Challenge (reasoning) [50], HellaSwag (commonsense) [51], MMLU (world knowledge) [52], and TruthfulQA (factuality) [53]. The models exhibit a wide variance in performance (e.g., MMLU scores ranging from 23.0 to 70.67), ensuring that our prediction task covers both low- and high-performance regimes.

For each model, we estimate its performance through distance-weighted $k$-NN interpolation ($k = 3$) based on the similarity of spectral signatures. We employ a 5-fold cross-validation scheme to ensure the generalization of our results. Additionally, we use EmbedLLM [1] as a strong activation-based baseline for performance quantification. Given a fixed prompt set $\mathcal{P}$, EmbedLLM encodes each model by extracting the hidden-state embedding at the final token position from the last transformer layer. A linear regression is trained on these embeddings to predict benchmark with leave-one-out algorithm.

We evaluate prediction quality using Mean Absolute Error (MAE) and Kendall's Tau rank correlation [54]. We use "Significance" as the number of repeated random train/test splits (out of 100) for which the Kendall's Tau correlation between predicted and ground-truth rankings is statistically significant at the 5% level. *Why spectral signatures are suitable here.* Benchmarking or activation-based embeddings require task data and inference, and do not scale well to frequent re-ranking across large model collections. Conversely, weight-only fingerprints such as PCS [11] and matrix-level verification methods such as AWM [20] are primarily designed for identity/provenance checks and do not naturally provide a compact, geometry-aware representation amenable to regression across heterogeneous architectures. Our spectral signatures provide a lightweight, task-agnostic representation that supports nearest-neighbor interpolation without any prompts.

*Results.* We run 100 repeated trials with random train/test splits (249 models for training and 250 models for testing in each trial). We report (i) the average MAE over the 100 trials, which measures absolute prediction error, and (ii) the number of trials in which Kendall's Tau correlation between predicted and ground-truth rankings on the test set is statistically significant. As shown in Tab. 6, across all benchmarks, spectral signatures achieve consistently low MAE and exhibit statistically significant rank correlation in all trials, indicating reliable prediction performance. While EmbedLLM achieves slightly lower MAE on ARC and TruthfulQA, spectral signatures outperform EmbedLLM on MMLU and remain competitive on HellaSwag. Notably, these results are obtained without using any task data or model inference, whereas EmbedLLM relies on data-driven embeddings. This comparison shows that compact and data-free spectral signatures can match the predictive accuracy of data-aware methods, offering a practical and scalable alternative for performance estimation in large LLM collections. Computationally, after one-time signature extraction per model, kNN prediction only requires distances in a small signature space rather than repeated forward passes over prompt sets.

*Computational Cost.* We report a direct runtime between our spectral signature and the activation-based baseline EmbedLLM under the same model and hardware settings. We use Llama-2-7b as the representative backbone and run both methods on a single NVIDIA L40 GPU with FP16 precision. For EmbedLLM, we follow the original pipeline and extract last-token hidden-state embeddings on a fixed prompt set. For our method, we compute the spectral signatures directly from model weights without any forward passes. We measure *Run Time* from the end of loading to the end of signature/embedding extraction. The Runtime of Spectral Signature is 337 seconds, while one iteration of EmbedLLM embeddings takes 2633 seconds.

## 5.4 Ablation Study

**PL_Alpha_Hill remains robust under output-invariant reparameterizations.** We evaluate whether PL_Alpha_Hill is a robust metric for constructing a spectral signature by performing output-invariant reparameterizations on weight matrices before measuring model similarity with Eq. 4. We compare the spectral signature constructed by PL_Alpha_Hill with other ESD scale metrics (Log Norm and Spectral Norm). Concretely, we apply two reparameterizations on LLaMA-2-7B model: (i) using output-invariant scaling to consecutive attention modules (Q/K and V/O pairs), and (ii) output-invariant change of basis between two consecutive linear layers in the feed-forward network (FFN) module (more details can be found in Appendix 7.8). Then, we compare our spectral similarity against PCS and other ESD-based scale metrics. Across scaling strengths, PCS degrades substantially, while PL_Alpha_Hill similarity remains essentially unchanged and stays near 1, correctly recognizing the transformed weights as functionally identical (Fig. 4). Furthermore, under MLP hidden-unit permutations, PCS collapses to near zero, but PL_Alpha_Hill remains near-perfect (Tab. 7).

**Table 5: KNN Interpolation Inverse Benchmark Results. Light gray rows indicate cosine correlation, and dark gray rows indicate sr (spearman rank) correlation.** [PR: move to appendix. compute significance (low priority)]

| Spectral Metric | Corr. | ARC | | HellaSwag | | MMLU | | TruthfulQA | | AVG | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | L2 | MAE | L2 | MAE | L2 | MAE | L2 | MAE | L2 |
| **Log_Alpha_Norm** | Cosine | 2.7214 | 5.0183 | 3.5390 | 8.1781 | 2.4005 | 5.0313 | 3.2106 | 4.5918 | 2.9679 | 5.7049 |
| | SR | **2.3631** | 4.2394 | 3.1378 | 7.1477 | **2.0581** | **3.9478** | 2.8943 | **3.9806** | 2.6133 | 4.8289 |
| **Spectral_Norm** | Cosine | 2.4795 | 4.3207 | 3.2681 | 7.3216 | 2.2043 | 4.2811 | 2.9508 | 4.2193 | 2.7257 | 5.0357 |
| | SR | 2.4608 | 4.2978 | 3.1511 | 7.0622 | 2.1499 | 4.0816 | 2.9752 | 4.2247 | 2.6843 | 4.9166 |
| **Stable_Rank** | Cosine | 2.3640 | **4.0669** | **2.9647** | **6.6716** | 2.1998 | 4.2833 | **2.8346** | 4.1694 | **2.5908** | **4.7978** |
| | SR | 2.4395 | 4.1770 | 3.1925 | 7.2703 | 2.1719 | 4.1112 | 2.9559 | 4.0674 | 2.6899 | 4.9065 |
| **PL_Alpha_Hill** | Cosine | 2.4679 | 4.3566 | 3.2265 | 7.3764 | 2.0640 | 4.0893 | 3.0459 | 4.3151 | 2.7011 | 5.0344 |
| | SR | 2.4581 | 4.3531 | 3.3280 | 7.6495 | 2.0972 | 4.0130 | 3.0366 | 4.2528 | 2.7300 | 5.0671 |

**Table 6: Performance prediction results on four benchmarks, comparing our spectral signature and EmbedLLM in terms of MAE and statistical significance.**

| | Spectral Signatures | | EmbedLLM | |
|---|---|---|---|---|
| **Benchmark** | **MAE** | **Significance** | **MAE** | **Significance** |
| ARC | 2.27 | 100 | 1.61 | 100 |
| HellaSwag | 3.25 | 100 | 3.50 | 100 |
| MMLU | 1.95 | 100 | 3.14 | 100 |
| TruthfulQA | 2.99 | 100 | 1.96 | 100 |

*Why this ablation matters.* Data-aware similarity (e.g., Logits and REEF) is expected to remain high under strictly output-invariant transforms, but it requires a prompt distribution and $O(\#\text{prompts})$ inference per comparison, making it unsuitable as a zoo-scale kernel. Weight-matrix fingerprints such as AWM [20] reduce some reparameterization sensitivity but still require matrix-level comparisons (and often assignment/matching) per pair. In contrast, our ESD *shape* signature is computed once per model and yields invariance to common output-preserving symmetries without explicit matching. Note that pure scale metrics (Log Norm, Spectral Norm) are also invariant here, but precisely because they track magnitude, they tend to saturate and are less discriminative across independently trained families (see Sec. 5.1).

**PL_Alpha_Hill remains robust under random perturbations.** To simulate post-training weight changes, we inject Gaussian noise of increasing magnitude into model weight entries and evaluate the stability of different similarity metrics. We follow the noise strength ($\gamma$) definition in Appendix 7.9 and report full results in Tab. 8. Across all noise levels, our PL_Alpha_Hill –based similarity remains close to the unperturbed model, but PCS degrades substantially as noise magnitude increases. These results indicate that spectral shape features provide a robust measure of model similarity under realistic weight-space perturbations. Moreover, spectral scale metrics also perform well in this scenario since additive Gaussian noise perturbs weights without strongly changing their overall magnitude. Beyond global averages, we analyze the layer-wise response to perturbation. As shown in Fig. 5, we observe that as noise intensity increases, PL_Alpha_Hill monotonically increases across all layers.

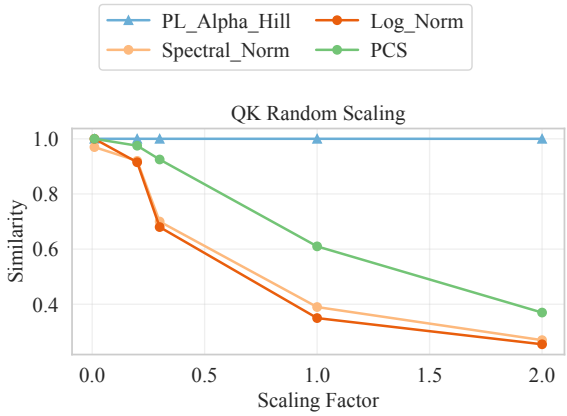



**Figure 4: Robustness of different similarity metrics under output-invariant random scaling. PL_Alpha_Hill is a shape metric, while Log Norm and Spectral Norm are scale metrics.**

**Table 7: Robustness of different similarity metrics under hidden-unit permutation.**

| Metric | MLP-In | MLP-Out |
|---|---|---|
| PCS | 0.00 | 0.00 |
| Log_Norm | 1.00 | 1.00 |
| Spectral_Norm | 1.00 | 1.00 |
| Log_Alpha_Norm | 1.00 | 1.00 |
| PL_Alpha_Hill (ours) | 1.00 | 1.00 |

## 6 Conclusions

This work presents a spectral shape–based framework for large-scale analysis of LLMs that is data-free, computationally efficient, and robust to post-training transformations. We curate and benchmark a large and diverse LLM corpus and conduct extensive experiments across similarity measurement, classifications, clustering,

**Table 8: Similarity scores of different metrics with increasing noise.**

| $\gamma$ | 0.05 | 0.10 | 0.20 | 0.30 | 0.50 | 1.00 | 2.00 |
|---|---|---|---|---|---|---|---|
| PCS | 1.00 | 1.00 | 0.98 | 0.96 | 0.89 | 0.70 | 0.44 |
| Log_Norm | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| Spectral_Norm | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.98 | 0.89 |
| Log_Alpha_Norm | 1.00 | 0.99 | 0.90 | 0.87 | 0.85 | 0.81 | 0.81 |
| PL_Alpha_Hill (ours) | 1.00 | 1.00 | 1.00 | 0.99 | 0.98 | 0.98 | 0.96 |

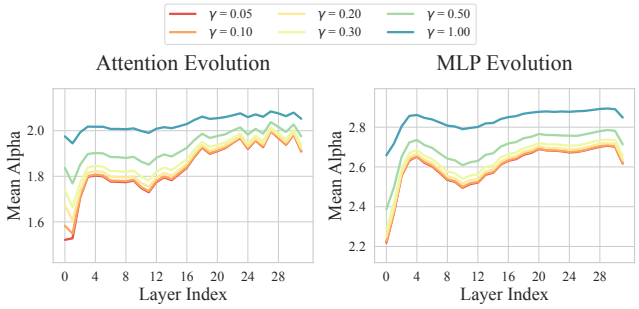



**Figure 5: PL_Alpha_Hill Evolution against Noise**

and performance prediction. Our results have shown the consistent effectiveness of our spectral signature across various experimental setups. Furthermore, we validate that ESD shape metrics, particularly PL_Alpha_Hill, capture intrinsic structural properties of LLMs that are stable under output-invariant reparameterizations and noise perturbations. Therefore, our spectral signature can provide a practical tool for managing modern LLM ecosystems, with implications for model provenance, organization, and scalable evaluation.

# References


[1] Richard Zhuang, Tianhao Wu, Zhaojin Wen, Andrew Li, Jiantao Jiao, and Kannan Ramchandran. Embedllm: Learning compact representations of large language models. In *ICLR*. OpenReview.net, 2025.

[2] Eliahu Horwitz, Nitzan Kurer, Jonathan Kahana, Liel Amar, and Yedid Hoshen. We should chart an atlas of all the world's models. *arXiv preprint arXiv:2503.10633v2*, 2025.

[3] Sally Zhu, Ahmed M. Ahmed, Rohith Kuditipudi, and Percy Liang. Independence tests for language models. In *ICML*. OpenReview.net, 2025.

[4] Eliahu Horwitz, Asaf Shul, and Yedid Hoshen. Unsupervised model tree heritage recovery. In *ICLR*. OpenReview.net, 2025.

[5] Charles H Martin and Michael W Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *Journal of Machine Learning Research*, 22(165):1–73, 2021.

[6] Charles H Martin, Tongsu Serena Peng, and Michael W Mahoney. Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data. *Nature Communications*, 12(1):1–13, 2021.

[7] Charles H Martin and Christopher Hinrichs. Setol: A semi-empirical theory of (deep) learning. *arXiv preprint arXiv:2507.17912*, 2025.

[8] Yaoqing Yang, Ryan Theisen, Liam Hodgkinson, Joseph E Gonzalez, Kannan Ramchandran, Charles H Martin, and Michael W Mahoney. Test accuracy vs. generalization gap: Model selection in nlp without accessing training or testing data. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3011–3021, 2023.

[9] Zhiguang Yang and Hanzhou Wu. A fingerprint for large language models. *ArXiv*, abs/2407.01235, 2024.

[10] Jie Zhang, Dongrui Liu, Chen Qian, Linfeng Zhang, Yong Liu, Yu Qiao, and Jing Shao. REEF: representation encoding fingerprints for large language models. In *ICLR*. OpenReview.net, 2025.

[11] Boyi Zeng, Lizheng Wang, Yuncong Hu, Yi Xu, Chenghu Zhou, Xinbing Wang, Yu Yu, and Zhouhan Lin. Huref: Human-readable fingerprint for large language models. In *Conference on Neural Information Processing Systems*, volume 37, pages 126332–126362, 2024.

[12] Ivica Nikolic, Teodora Baluta, and Prateek Saxena. Model provenance testing for large language models. In *Advances in Neural Information Processing Systems*, 2025.

[13] Heng Jin, Chaoyu Zhang, Shanghao Shi, Wenjing Lou, and Y Thomas Hou. Proflingo: A fingerprinting-based intellectual property protection scheme for large language models. In *2024 IEEE Conference on Communications and Network Security (CNS)*, pages 1–9. IEEE, 2024.

[14] Mingjie Sun, Yida Yin, Zhiqiu Xu, J Zico Kolter, and Zhuang Liu. Idiosyncrasies in large language models. *arXiv preprint arXiv:2502.12150*, 2025.

[15] Kangxi Wu, Liang Pang, Huawei Shen, Xueqi Cheng, and Tat-Seng Chua. Llmdet: A third party large language models generated text detection tool. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

[16] Tianhao Wang and Florian Kerschbaum. Attacks on digital watermarks for deep neural networks. In *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019.

[17] Yusuke Uchida, Yuki Nagai, Shigeyuki Sakazawa, and Shin'ichi Satoh. Embedding watermarks into deep neural networks. New York, NY, USA, 2017. Association for Computing Machinery.

[18] Chenxi Gu, Chengsong Huang, Xiaoqing Zheng, Kai-Wei Chang, and Cho-Jui Hsieh. Watermarking pre-trained language models with backdooring. *ArXiv*, abs/2210.07543, 2022.

[19] Ya Jiang, Chuxiong Wu, Massieh Kordi Boroujeny, Brian Mark, and Kai Zeng. Stealthink: A multi-bit and stealthy watermark for large language models. In *Forty-second International Conference on Machine Learning*, 2025.

[20] Boyi Zeng, Lin Chen, Ziwei He, Xinbing Wang, and Zhouhan Lin. Awm: Accurate weight-matrix fingerprint for large language models. *arXiv preprint arXiv:2510.06738*, 2025.

[21] Suqing Wang, Ziyang Ma, Xinyi Li, and Zuchao Li. Ghost in the transformer: Tracing llm lineage with svd-fingerprint. *arXiv preprint arXiv:2511.06390*, 2025.

[22] Do-hyeon Yoon, Minsoo Chun, Thomas Allen, Hans Müller, Min Wang, and Rajesh Sharma. Intrinsic fingerprint of llms: Continue training is not all you need to steal a model! *arXiv preprint arXiv:2507.03014*, 2025.

[23] Almog Gueta, Elad Venezian, Colin Raffel, Noam Slonim, Yoav Katz, and Leshem Choshen. Knowledge is a region in weight space for finetuned language models. *arXiv preprint arXiv:2302.04863v3*, 2023.

[24] Konstantin Schürholt, Léo Meynent, Yefan Zhou, Haiquan Lu, Yaoqing Yang, and Damian Borth. A model zoo on phase transitions in neural networks. *arXiv preprint arXiv:2504.18072v1*, 2025.

[25] Yaoqing Yang, Liam Hodgkinson, Ryan Theisen, Joe Zou, Joseph E Gonzalez, Kannan Ramchandran, and Michael W Mahoney. Taxonomizing local versus global structure in neural network loss landscapes. In *In Advances in Neural Information Processing Systems*, volume 34, pages 18722–18733. Curran Associates, Inc., 2021.

[26] Michael Mahoney and Charles Martin. Traditional and heavy tailed self regularization in neural network models. In *International Conference on Machine Learning*, pages 4284–4293. PMLR, 2019.

[27] Romain Couillet and Zhenyu Liao. *Random matrix methods for machine learning*. Cambridge University Press, 2022.

[28] Zhenyu Liao and Michael W Mahoney. Random matrix theory for deep learning: Beyond eigenvalues of linear models. *arXiv preprint arXiv:2506.13139*, 2025.

[29] Mert Gurbuzbalaban, Umut Simsekli, and Lingjiong Zhu. The heavy-tail phenomenon in sgd. In *International Conference on Machine Learning*, pages 3964–3975. PMLR, 2021.

[30] Liam Hodgkinson and Michael Mahoney. Multiplicative noise and heavy tails in stochastic optimization. In *International Conference on Machine Learning*, pages 4262–4274. PMLR, 2021.

[31] Liam Hodgkinson, Umut Simsekli, Rajiv Khanna, and Michael Mahoney. Generalization bounds using lower tail exponents in stochastic optimizers. In *International Conference on Machine Learning*, pages 8774–8795. PMLR, 2022.

[32] Umut Simsekli, Levent Sagun, and Mert Gurbuzbalaban. A tail-index analysis of stochastic gradient noise in deep neural networks. In *International Conference on Machine Learning*, pages 5827–5837. PMLR, 2019.

[33] Umut Simsekli, Ozan Sener, George Deligiannidis, and Murat A Erdogdu. Hausdorff dimension, heavy tails, and generalization in neural networks. *Advances in Neural Information Processing Systems*, 33:5138–5151, 2020.

[34] Yaoqing Yang, Ryan Theisen, Liam Hodgkinson, Joseph E Gonzalez, Kannan Ramchandran, Charles H Martin, and Michael W Mahoney. Test accuracy vs. generalization gap: Model selection in nlp without accessing training or testing data. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3011–3021, 2023.

[35] Yefan Zhou, Tianyu Pang, Keqin Liu, Michael W Mahoney, Yaoqing Yang, et al. Temperature balancing, layer-wise weight analysis, and neural network training. *Advances in Neural Information Processing Systems*, 36:63542–63572, 2023.

[36] Zihang Liu, Yuanzhe Hu, Tianyu Pang, Yefan Zhou, Pu Ren, and Yaoqing Yang. Model balancing helps low-data training and fine-tuning. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1311–1331, Miami, Florida, USA, November 2024. Association for Computational Linguistics.

[37] Yuanzhe Hu, Kinshuk Goel, Vlad Killiakov, and Yaoqing Yang. Eigenspectrum analysis of neural networks without aspect ratio bias. In *ICML*. OpenReview.net, 2025.

[38] Haiquan Lu, Yefan Zhou, Shiwei Liu, Zhangyang Wang, Michael W Mahoney, and Yaoqing Yang. Alphapruning: Using heavy-tailed self regularization theory for improved layer-wise pruning of large language models. *Advances in neural information processing systems*, 37:9117–9152, 2024.

[39] Bruce M Hill. A simple general approach to inference about the tail of a distribution. *The annals of statistics*, pages 1163–1174, 1975.

[40] Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models. 2023.

[41] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 3214–3252, 2022.

[42] Hugo Touvron, Thibaut Lavril, Gautier Izacard, et al. Llama: Open and efficient foundation language models. 2023.

[43] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The llama 3 herd of models. 2024.

[44] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, et al. Mistral 7b. 2023.

[45] Xinyang Geng and Hao Liu. Openllama: An open reproduction of llama, May 2023.

[46] Jinze Bai, Shuai Bai, Yunfei Chu, et al. Qwen technical report, 2023.

[47] Peter J Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20:53–65, 1987.

[48] David L Davies and Donald W Bouldin. A cluster separation measure. *IEEE transactions on pattern analysis and machine intelligence*, (2):224–227, 2009.

[49] Aidar Myrzakhan, Sondos Mahmoud Bsharat, and Zhiqiang Shen. Open-llm-leaderboard: From multi-choice to open-style questions for llms evaluation, benchmark, and arena. *arXiv preprint arXiv:2406.07545*, 2024.

[50] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv*, 2018.

[51] Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a machine really finish your sentence? In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4791–4800, Florence, Italy, July 2019. Association for Computational Linguistics.

[52] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021.

[53] Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252. Association for Computational Linguistics, 2022.
[54] Maurice George Kendall. Rank correlation methods. 1948.

# 7 Appendix

## 7.1 Metric Computation

In this section, we provide the detailed definitions of the metrics used in our analysis. Let $\mathbf{W}$ represent the weight matrix.

*Log_Norm.* Log_Norm is defined as the logarithm of the squared Frobenius norms:

$$\text{Log_Norm} = \log \|\mathbf{W}\|_F^2. \tag{8}$$

*Log_Alpha_Norm.* This metric is a scale-adjusted metric based on the Schatten $p$-norm. Let $\{\lambda_j\}_{j=1}^{M}$ denote the set of eigenvalues of the correlation matrix $\mathbf{X} = \mathbf{W}^\top \mathbf{W}$, where $\mathbf{W}$ is an $N \times M$ weight matrix (assuming $N \geq M$). The Schatten $p$-norm is defined as:

$$\|\mathbf{X}\|_p = \left( \sum_{j=1}^{M} \lambda_j^p \right)^{\frac{1}{p}}. \tag{9}$$

The Log_Alpha_Norm metric is then given by:

$$\text{Log_Alpha_Norm} = \log \|\mathbf{X}\|_\alpha^\alpha. \tag{10}$$

*Spectral_Norm.* The Spectral_Norm focuses on the spectral properties of the weight matrices. Unlike the logarithmic version, here we compute the average squared spectral norm (operator norm) across all layers:

$$\text{Spectral_Norm} = \|\mathbf{W}\|_2^2, \tag{11}$$

where $\|\mathbf{W}\|_2$ denotes the spectral norm of the matrix $\mathbf{W}$, which corresponds to its largest singular value.

## 7.2 Metrics for Classification Quality

Given two signatures $\mathbf{Z}_1, \mathbf{Z}_2$, we consider: (i) Spearman rank correlation $\rho_s(\mathbf{Z}_1, \mathbf{Z}_2)$ as our default similarity (Eq. 5), (ii) an RBF kernel applied to the signature distance (Eq. 12),

$$k_{\text{RBF}}(\mathbf{Z}_1, \mathbf{Z}_2) = \exp\left( -\frac{\|\mathbf{Z}_1 - \mathbf{Z}_2\|_2^2}{2\sigma^2} \right) \tag{12}$$

, and (iii) cosine similarity (Eq. 13).

$$k_{\cos}(Z_1, \mathbf{Z}_2) = \frac{\langle \mathbf{Z}_1, \mathbf{Z}_2 \rangle}{\|\mathbf{Z}_1\|_2 \|\mathbf{Z}_2\|_2} \tag{13}$$

## 7.3 Metrics for Internal Clustering Validation

In the absence of ground truth labels for the unsupervised discovery of model families, we rely on internal validity indices to evaluate the quality of the clustering structures derived from spectral signatures. We employ two standard metrics: the Silhouette Coefficient and the Davies-Bouldin Index. Detailed definitions are provided below.

*Silhouette Coefficient .* The Silhouette Coefficient evaluates the quality of clustering by measuring how similar an object is to its own cluster (cohesion) compared to other clusters (separation). For a data point $i \in C_k$ (where $C_k$ is a given cluster), let $a(i)$ be the average distance between $i$ and all other points in the same cluster, and let $b(i)$ be the minimum average distance from $i$ to all points in any other cluster $C_j$ (where $j \neq k$). The silhouette coefficient $s(i)$ is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{14}$$

The overall Silhouette Score is the mean of $s(i)$ across all samples.

*Davies-Bouldin Index.* The Davies-Bouldin Index (DBI) signifies the average 'similarity' between clusters, where similarity is a measure that compares the distance between clusters with the size of the clusters themselves.

Let $S_i$ be the intra-cluster dispersion (e.g., average distance of points to the centroid) of cluster $i$, and let $M_{ij}$ be the distance between the centroids of clusters $i$ and $j$. The similarity ratio $R_{ij}$ is defined as:

$$R_{ij} = \frac{S_i + S_j}{M_{ij}} \tag{15}$$

The Davies-Bouldin Index is defined as the average of the maximum similarity ratios for each cluster:

$$DB = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} R_{ij} \tag{16}$$

## 7.4 Similarity within Modules

Fig. 6 shows module-wise Spearman similarity heatmaps computed from layer-wise PL_Alpha_Hill profiles. A clear contrast emerges between attention and MLP modules. For the attention Q projection, many cross-family pairs remain highly similar (e.g., open-llama-7b vs. llama-2-7b reaches Sim $\approx 0.90$), resulting in a relatively weak separation between related and unrelated models. In contrast, the MLP up projection module exhibits a much sharper block-wise structure, where models derived from the same base family form high-similarity blocks, while similarities across families drop substantially.

We hypothesize that this difference reflects the roles of the two components in Transformer weights. Attention projections are strongly constrained by shared architectural parameterization and generic token-mixing behavior, which can lead to similar spectral shapes even across independently trained models of comparable scale. MLP projections, however, are more directly responsible for feature transformation and information storage, and thus may encode more lineage-specific training signals, producing clearer family-level clusters in spectral space.

For a model pool with $n$ models, we can derive a similarity matrix $S \in \mathbb{R}^{n \times n}$. To provide a concrete illustration of the spectral signature, we present a representative pair of models in Fig. **??**.

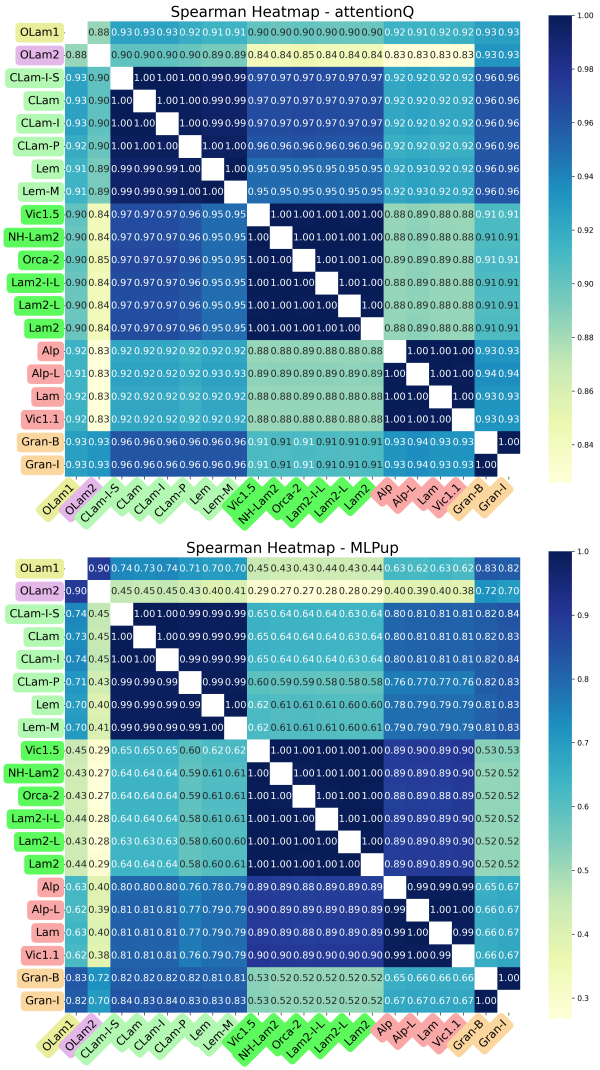



**Figure 6: Module-wise spectral similarity heatmaps. PL_Alpha_Hill-SR signatures computed separately for (top) the attention Q projection and (bottom) the MLP projection module.**

## 7.5 Additional Details on the Robustness of ESD Shape Metrics

Pre-training and post-training processes inherently reshape the correlations within weight matrices, which are reflected in the variations of the heavy-tail index across layers. To validate the reliability of our method, we investigate whether the spectral structure preserves sufficient lineage information despite these post-training perturbations.

*PL_Alpha_Hill is robust under perturbation.* Fig. 7 shows that ESD shape metrics can encode lineage information beyond pretraining. Using Spearman correlation between layer-wise PL_Alpha_Hill profiles, models derived from the same 7B base checkpoint form

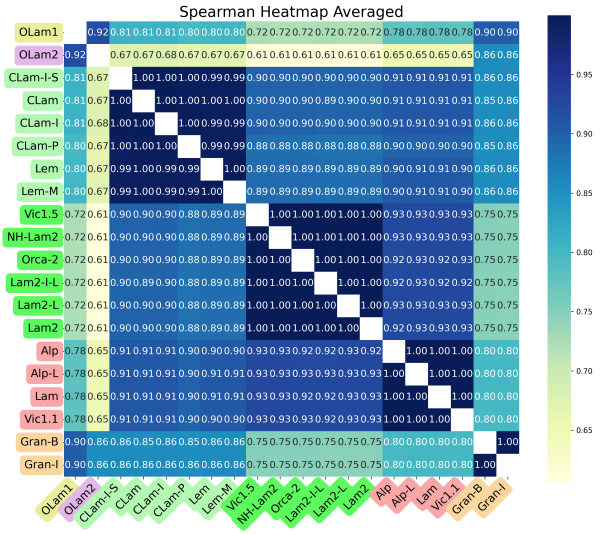



**Figure 7: Heatmap of spectral signature similarity across models derived from different 7B base models. The pairwise similarity is measured by Spearman's rank correlation of PL_Alpha_Hill. Entries are color-coded on the axes to indicate the base model. Specifically, different shades of green represent models derived from Llama-2-7b. While dark green denotes the ones directly fine-tuned from the Llama-2-7B family, light green identifies the CodeLlama-7B-hf sub-family. The emergence of distinct clusters aligned with these color markers demonstrates that spectral signatures are highly sensitive to ancestral relations, even after diverse fine-tuning trajectories. We provide the full name list for the abbreviations in Tab. 9.**

clear high-similarity blocks, whereas similarities across base families are substantially lower. Moreover, within the Llama-2 lineage, we observe finer sub-structure: distinct clusters align with different post-training trajectories (e.g., instruction-tuned vs. CodeLlama-style derivatives), suggesting that post-training alters spectral signatures in a structured manner. These observations motivate our use of spectral signatures for downstream tasks such as lineage attribution, clustering, and performance prediction (Sec.5).

*The Forging of Spectral Signature.* As illustrated in Fig. 8, we observe three distinct phases of spectral evolution:

(1) **Initialization.** At Step 0, all layers exhibit large, uniform PL_Alpha_Hill values ($\alpha \approx 2.1$ for attention weight matrices and $\alpha \approx 3.5$ for MLP weight matrices, due to the aspect ratio of weight matrices [37]), resembling random matrices with no structural identity.
(2) **Formulation.** As training progresses, $\alpha$ decreases globally, signifying the accumulation of information. Crucially, a distinct layer-wise topology emerges: PL_Alpha_Hill drop significantly in early layers (feature extraction), flatten in middle layers, dip in mid-deep layers, and rise in the final layers.

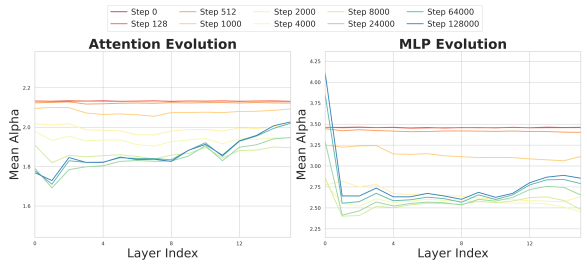



**Figure 8: The evolution of layer-wise PL_Alpha_Hill through the training process. The values of PL_Alpha_Hill are averaged within Attention modules and MLP modules.**

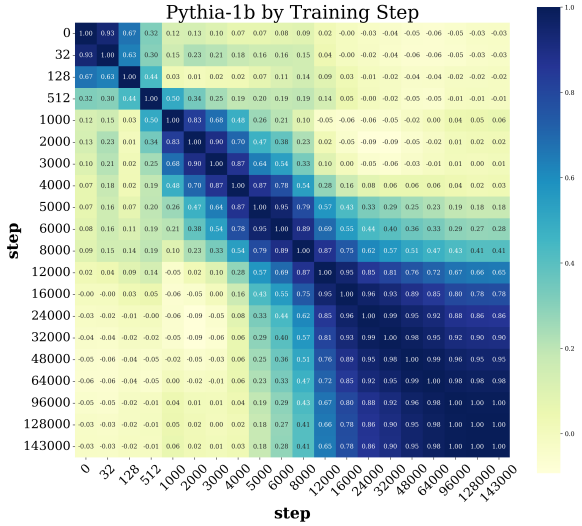



**Figure 9: The heatmap of the Spearman rank between PL_Alpha_Hill signatures of pythia-1b models with training steps ranging from step 0 to step 143,000.**

(3) **Stabilization** As shown in the temporal similarity heatmap (Fig. 9), the spectral signature stabilizes rapidly. While the early phase ($< 16k$ steps) shows high volatility, the signature "locks in" after step 16,000. Subsequent updates (from 16k to 143k) refine the weights but do not alter the fundamental topological shape.

*PL_Alpha_Hill undergo minimal change during post-training.* Because post-training updates (e.g., supervised fine-tuning and reinforcement learning from human feedback) typically introduce relatively small changes in parameter space, they are unlikely to erase the heavy-tailed structure established during pre-training. Based on that, PL_Alpha_Hill is promising to work as a stable and lineage-preserving signature: it is sensitive to the underlying architecture and pre-training data, but remains robust to common downstream adaptations and moderate noise. This motivates our use of ESD-based metrics for both performance prediction and model attribution.

**Table 9: Mapping of Model Abbreviations to Full Model Names**

| Abbreviation | Full Model Name |
|---|---|
| OLam1 | openlm-research/open_llama_7b |
| OLam2 | openlm-research/open_llama_7b_v2 |
| CLam-I-S | AlfredPros/CodeLlama-7b-Instruct-Solidity |
| CLam | codellama/CodeLlama-7b-hf |
| CLam-I | codellama/CodeLlama-7b-Instruct-hf |
| CLam-P | codellama/CodeLlama-7b-Python-hf |
| Lem | EleutherAI/llemma_7b |
| Lem-M | EleutherAI/llemma_7b_muinstruct_camelmath |
| Vic1.5 | lmsys/vicuna-7b-v1.5 |
| NH-Lam2 | NousResearch/Nous-Hermes-llama-2-7b |
| Orca-2 | microsoft/Orca-2-7b |
| Lam2-I-L | lvkaokao/llama2-7b-hf-instruction-lora |
| Lam2-L | oh-yeontaek/llama-2-7B-LoRA-assemble |
| Lam2 | meta-llama/Llama-2-7b-hf |
| Alp | wxjiao/alpaca-7b |
| Alp-L | chainyo/alpaca-lora-7b |
| Lam | luodian/llama-7b-hf |
| Vic1.1 | lmsys/vicuna-7b-v1.1 |
| Gran-B | ibm-granite/granite-7b-base |
| Gran-I | ibm-granite/granite-7b-instruct |

## 7.6 Additional Results on Model Clustering

In Fig. 10, we show a t-SNE visualization of a 2D embedding for spectral clustering on PL_Alpha_Hill with ground truth cluster labels.

## 7.7 Additional Results on Performance Predictions

Fig. 11 compares predicted benchmark scores with their ground-truth values. The points cluster closely around the diagonal ($y = x$), indicating good agreement between predicted and actual performance.

## 7.8 Additional Details on Robustness to Output-invariant Reparameterization.

We first evaluate whether our spectral signature PL_Alpha_Hill is stable under output-invariant reparameterizing transformations that preserve the model's input-output function for all inputs.

Consider a standard Transformer layer with self-attention and feed-forward networks (FFN):

$$H'_{attn} = softmax(\frac{H_n W_Q (H_n W_K)^T}{\sqrt{d}}) H_n W_V W_O \tag{17}$$

$$H'_{n+1} = \text{FFN}(H'_{attn}) = \sigma(H'_{attn} W_1 + b_1) W_2 + b_2 \tag{18}$$

Based on these structures, we consider two representative classes of reparameterizations.

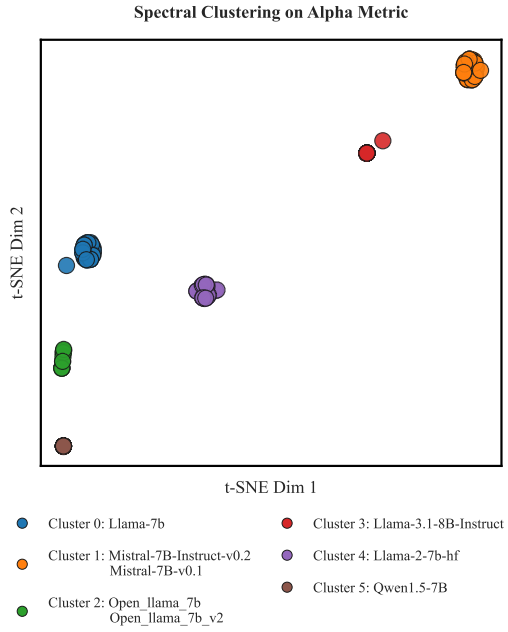



**Figure 10: t-SNE 2D embedding for spectral clustering on PL_Alpha_Hill with ground truth cluster labels.**

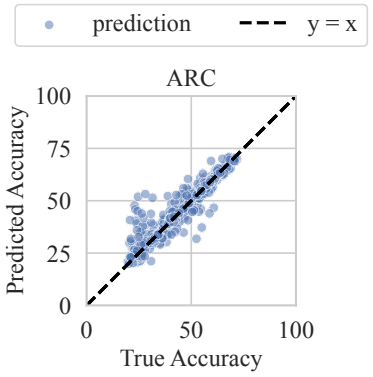



**Figure 11: A scatter plot of predicted and ground-truth ARC scores across the model pool.**

**Linear Chains.** Consider a sequence of linear maps $W_1, W_2, \ldots, W_L$ without introducing non-linearality. Scaling each matrix as:

$$W_i' = c_i W_i \tag{19}$$

The overall linear map would be preserved as long as $\prod_{i=1}^{L} c_i = 1$. In a Transformer layer, we notice that $W_Q, W_K$ and $W_V, W_O$ form two such linear pairs in Eq. 17. We apply balanced scaling to each pair so that the attention output remains unchanged. The PL_Alpha_Hill is derived from the ESD of the Gram matrix, which depends on the distribution of eigenvalues rather than their absolute magnitude. As a result, PL_Alpha_Hill is invariant to balanced re-scaling within linear chains and remains stable under such output-invariant scaling.

**Permutation on Hidden Units.** Due to the symmetry of fully-connected layers, the hidden units of a two-layer FFN can be permuted without changing the input-output function, as long as the permutation is applied consistently. Let $P$ be a permutation matrix. Eq. 18 remains unchanged under:

$$W_1' = PW_1, \quad W_2' = W_2 P^{-1}, \quad b_1' = Pb_1.$$

This transformation corresponds to a similarity transform on the associated Gram matrices. Since permutation matrices are orthogonal, i.e., $P^{-1} = P^{\top}$, the singular values of the Gram matrix, and hence its ESD remain identical. As a result, all ESD-based metrics, including PL_Alpha_Hill, are exactly invariant under hidden-unit permutations. We note that this setting differs from post-hoc neuron matching approaches (e.g., via the Hungarian algorithm [3]), as the permutation considered here preserves the model function by construction.

*Experimental Setup.* We conduct our analysis on the pre-trained Llama-2-7b model using the following reparameterization formulations.

**Random Scaling.** To simulate a "disguised" linear chain, we multiply the weight matrix pairs by random numbers, as defined in Eq. 19. We draw a log-scale value $z \sim \mathcal{N}(0,1)$ and define a scaling factor $a = \exp(\gamma z)$ controlled by strength $\gamma$. The weights are updated as $Q' = aQ, K' = K/a$ (and similarly for $V, O$).

**Permutation on Hidden Units.** For each layer, we generate a random permutation matrix $P$ and construct a transformed weight set

$$W' = \{PW_{gate}, PW_{up}, W_{down}P^{-1}\}$$

*Results.* We compare two similarity metrics between the original weights $W$ and transformed weights $W'$.

**Naive Metric:** Layer-wise Cosine Similarity, which assumes coordinate alignment. Specifically, we use the resemblance definition of similarity in [11]. The weight matrix of each layer is viewed as a vector and then concatenated. We compute the cosine similarity between two giant vectors from two models.

**Spectral Metric:** The difference in the spectral signature. Specifically, for each module (e.g, attention Q), we extract the layer-wise spectral metrics. The resulting metrics are treated as a vector with $L$ entries. The similarity of a module between two models are defined by the vector cosine similarity. The similarities of such module pairs are then averaged as the final similarity between models. In our setting, we select `PL_Alpha_Hill, norm, log norm, spectral norm` as the spectral metrics.

**Invariant Terms:** To mitigate the effect of output-invariant reparameterizations, we introduce the invariant terms. Specifically, we can utilize the symmetric attribution of the transformer networks by multiplying such weight matrices. For the linear chains, we introduce:

$$M_a = W_Q W_K \quad \text{or} \quad M_a = W_V W_O.$$

For the permuted hidden units, we introduce:

$$M_b = W_1 W_2.$$

## 7.9 Additional Details on Stability under Random Perturbations

To rigorously evaluate the stability of our spectral signatures (PL_Alpha_Hill), we apply controlled random perturbations to pre-trained weights. This experiment is designed to simulate the post-training process, assessing whether the heavy-tailed signature is a robust topological feature or a fragile artifact of specific floating-point parameter values.

*Setup.* For every weight matrix $W_\ell$ in the model, we first compute its entry-wise standard deviation $\mathrm{std}(W_\ell)$, then sample a standard Gaussian matrix $Z_\ell \sim \mathcal{N}(0, I)$ of the same shape. We define the unit perturbation as

$$\Delta W_\ell = \mathrm{std}(W_\ell)\, Z_\ell, \tag{20}$$

and obtain a noise-perturbed model at scale $\gamma$ by

$$W_\ell^{(\gamma)} = W_\ell + \gamma \Delta W_\ell, \tag{21}$$

where $\gamma$ controls the relative noise strength. We pre-generate and store $\{\Delta W_\ell\}_\ell$ once, and reuse the same unit adapter across different $\gamma$ values to enable controlled comparisons.